

## White Paper

### Phish Shield: An Automated Phishing Email Detection System

#### Executive Summary

Phishing emails remain one of the most common and damaging cyber threats, exploiting human trust to steal sensitive information and deploy malware. This white paper presents Phish Shield, an automated phishing email detection system that combines Natural Language Processing (NLP), machine learning, and rule-based analysis to detect malicious emails in real time.

#### 1. Problem Overview

Phishing attacks impersonate trusted entities to deceive users. Traditional filters fail to detect modern phishing methods, creating the need for intelligent automated solutions.

#### 2. Limitations of Existing Solutions

Existing systems often rely only on text analysis, lack semantic understanding, and struggle with real-time performance and data imbalance.

#### 3. Proposed Solution: Phish Shield

Phish Shield is a hybrid detection system analyzing headers, content, URLs, and attachments using a modular Chain of Responsibility architecture.

#### 4. System Architecture

Email Input → Header Analysis → URL Analysis → Content NLP → ML Classification → Alert or Quarantine

#### 5. Methodology

Datasets were analyzed using TF-IDF features, NLTK preprocessing, and ML models including Logistic Regression, SVM, Random Forest, and an Ensemble model.

#### 6. Evaluation and Results

The ensemble model achieved high accuracy and robustness, outperforming individual classifiers.

#### 7. Practical Use Case

Phish Shield can be deployed as an organizational email gateway to automatically detect and quarantine phishing emails.

#### 8. Security and Ethical Considerations

User privacy is preserved, false positives are minimized, and data is processed securely.

## 9. Conclusion

Phish Shield provides a practical, scalable solution for phishing email detection with strong performance and future extensibility.